

Form Number : _____

Paper Code
(0000CMD300118003)

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**CLASSROOM CONTACT PROGRAMME**
(Academic Session : 2018 - 2019)**PRE-MEDICAL : NURTURE COURSE**
ALL PHASE

Test Type : ALL INDIA OPEN TEST

Test Pattern : NEET(UG)

TEST DATE : 17 - 02 - 2019

TEST SYLLABUS : FULL SYLLABUS

Important Instructions

Do not open this Test Booklet until you are asked to do so

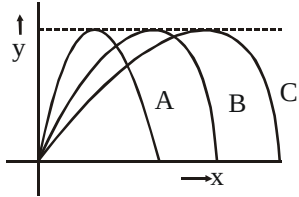
1. A seat marked with Reg. No. will be allotted to each student. The student should ensure that he/she occupies the correct seat only.
If any student is found to have occupied the seat of another student, both the students shall be removed from the examination and shall have to accept any other penalty imposed upon them.
2. Duration of Test is 3 Hours and Questions Paper Contains 180 Questions. The Max. Marks are 720.
3. Student can not use log tables and calculators or any other material in the examination hall.
4. Student must abide by the instructions issued during the examination, by the invigilators or the centre incharge.
5. Before attempting the question paper ensure that it contains all the pages and that no question is missing.
6. Each correct answer carries 4 marks, while 1 mark will be deducted for every wrong answer. Guessing of answer is harmful.
7. A candidate has to write his / her answers in the OMR sheet by darkening the appropriate bubble with the help of Blue / Black Ball Point Pen only as the correct answer(s) of the question attempted.
8. Use of Pencil is strictly prohibited.

Note : In case of any Correction in the test paper, please mail to dlpcorrections@allen.ac.in within 2 days along with Paper code and Your Form No.

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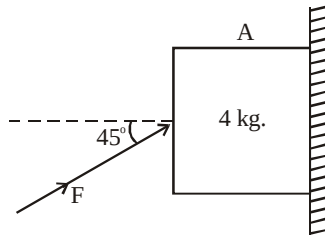
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BEWARE OF NEGATIVE MARKING

- If $y = \sin\theta + \sqrt{3}\cos\theta$, then range of y is :-
 (1) $0 \leq y \leq 2$ (2) $-2 \leq y \leq 0$
 (3) $-2 \leq y \leq 2$ (4) $0 \leq y \leq 4$
- Given that $\vec{A} + \vec{B} + \vec{C} = \vec{0}$. Out of these three vectors two are equal in magnitude and the magnitude of the third vector is $\sqrt{2}$ times that of either of the two having equal magnitude. Then the angles between vectors are :-
 (1) $30^\circ, 60^\circ, 90^\circ$ (2) $45^\circ, 45^\circ, 90^\circ$
 (3) $45^\circ, 60^\circ, 90^\circ$ (4) $90^\circ, 135^\circ, 135^\circ$
- If three vectors satisfy the relation $\vec{A} \cdot \vec{B} = 0$ and $\vec{A} \cdot \vec{C} = 0$, the $\vec{A}$ can be parallel to vector :-
 (1) $\vec{C}$ (2) $\vec{B}$ (3) $\vec{B} \times \vec{C}$ (4) $\vec{B} \cdot \vec{C}$
- If A, B and C are physical quantities, having different dimensions, which of the following combinations can never be a meaningful quantity ?
 (i) $\frac{AB}{C}$ (ii) $\frac{A+B}{C}$
 (iii) $AB-C^2$ (iv) $(A-B)C$
 (1) (iii, iv) (2) (i, ii)
 (3) (ii, iv) (4) (ii, iii)
- You measure two quantities as $A = (1.0 \pm 0.2)\text{m}$ and $B = (2.0 \pm 0.2)\text{m}$. You should report correct value for $\sqrt{AB}$ as :-
 (1) $(1.4 \pm 0.4)\text{m}$
 (2) $(1.41 \pm 0.15)\text{m}$
 (3) $(1.4 \pm 0.3)\text{m}$
 (4) $(1.4 \pm 0.2)\text{m}$
- A student stands at the edge of a cliff 50 m above a beach and throws a stone horizontally over the edge with a speed of 18 m/s. How long after being released does the stone strike the beach below the cliff ? (Take $g = 10 \text{ m/s}^2$)
 (1) 10 s (2) 5 s
 (3) $\sqrt{5}\text{s}$ (4) $\sqrt{10}\text{s}$
- In Q. no 6 with what speed does the stone land ?
 (1) 36.4 m/s (2) 18.0 m/s
 (3) 9.0 m/s (4) $\sqrt{10}\text{ m/s}$
- In Q. no. 6 at which angle does the stone land ?
 (1) $\tan^{-1}\left(\frac{-10}{18}\right)$ (2) $\tan^{-1}\left(\frac{-18}{10}\right)$
 (3) $\tan^{-1}\left(\frac{-10\sqrt{10}}{18}\right)$ (4) $\tan^{-1}\left(\frac{-18}{10\sqrt{10}}\right)$
- Two particles are projected from the same point on the ground with same speed simultaneously at angles θ_1 and θ_2 with the horizontal. They strike the ground at same point after t_1 and t_2 respectively, then :
 (1) $\frac{t_1}{t_2} = \frac{\sin\theta_1}{\sin\theta_2}$ (2) $\frac{t_1}{t_2} = \tan\theta_1$
 (3) $\frac{t_1}{t_2} = 1$ (4) All of these
- Three projectiles A, B and C are thrown from the same point in the same plane. Their trajectories are shown in the figure. Which among the following statement is correct?

 (1) The projectile speed is greater for particle C.
 (2) The time of flight is the same for all the three
 (3) The horizontal component of velocity is greater for particle C
 (4) All of the above
- If impulse (In $\text{kg} \cdot \text{ms}^{-1}$) varies with time t as $20t^2 - 40t$. Then the change in momentum is minimum at time :-
 (1) $t = 2\text{ s}$ (2) $t = 1\text{ s}$
 (3) $t = \frac{1}{2}\text{ s}$ (4) $t = \frac{3}{2}\text{ s}$

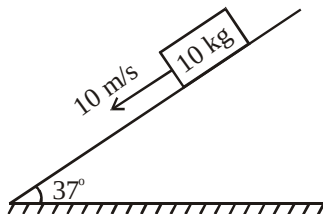
प्रत्येक प्रश्न को अर्जुन बनकर करो।

12. Block A of mass 4 kg is to be kept at rest against a smooth vertical wall by applying a force F as shown in figure. The force required is ($g = 10 \text{ m/s}^2$) :-



- (1) $40\sqrt{2} \text{ N}$ (2) $20\sqrt{2} \text{ N}$
(3) $10\sqrt{2} \text{ N}$ (4) $15\sqrt{2} \text{ N}$

13. Block of mass 10 kg is moving on inclined plane with constant velocity 10 m/s as shown in figure. The coefficient of kinetic friction between incline plane and block is :-



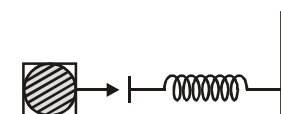
- (1) 0.57 (2) 0.5 (3) 0.75 (4) 0.90

14. A man of mass m is standing in an elevator moving downward with an acceleration $\frac{g}{5}$. The force exerted by the bottom surface of the elevator on the man will be :-

- (1) $\frac{4mg}{5}$ (2) $\frac{6mg}{5}$ (3) $\frac{2mg}{5}$ (4) $\frac{mg}{5}$

15. The kinetic energy of a body is increased by 300%. The momentum of the body would increase by :
(1) 50% (2) 100%
(3) 150% (4) 300%

16. A mass of 0.5 kg moving with a speed of 1.5 m/s on a horizontal smooth surface, collides with a nearly weightless spring of force constant 50 N/m. The maximum compression of the spring would be :

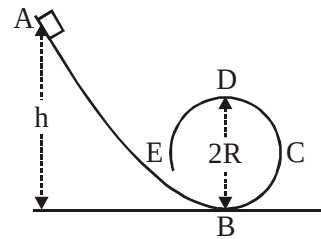


- (1) 0.15m (2) 0.12m
(3) 1.5m (4) 0.5m

17. An athlete of mass 60 kg jumps at the rate of 20 steps per minute through an average height of 25cm. The power developed is :

- (1) 98 W (2) 49W (3) 14W (4) 21W

18. A frictionless track ABCDE ends in a circular loop of radius R. A body slides down the track from point A which is at a height $h = 5 \text{ cm}$. Maximum value of R for the body to successfully complete the loop is :



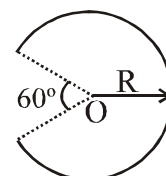
- (1) 5cm (2) $\frac{15}{4} \text{ cm}$ (3) $\frac{10}{3} \text{ cm}$ (4) 2cm

19. A toy-cart is tied to the end of an unstretched elastic string of length "a". When revolved, the toy-cart moves in a horizontal circle of radius "2a" with a time period T_1 . Now the toy-cart is speeded until it moves in a horizontal circle of radius "3a" with a period T_2 , then

- (1) $T_2 = T_1$ (2) $T_2 = \frac{3}{2} T_1$

- (3) $T_2 = \frac{\sqrt{3}}{2} T_1$ (4) $T_2 = \frac{T_1}{2}$

20. A wire is bent in the form of a circular arc as shown in fig. The distance of centre of mass of the system from point O is -



- (1) $\frac{5R}{3\pi}$ (2) $\frac{3\sqrt{3}R}{\pi}$

- (3) $\frac{3R}{5\pi}$ (4) $\frac{3R}{\pi}$

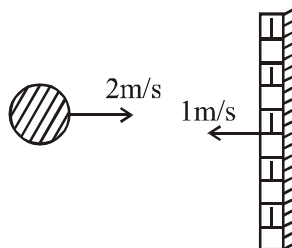
21. Two particles of masses m_1 and m_2 are projected from the top of a tower. The particle m_1 is projected vertically downward with speed u and particle m_2 is projected horizontally with same speed. The acceleration of centre of mass of the system is - (neglect air resistance)

- (1) g , downward
(2) $\frac{m_1 g}{m_1 + m_2}$, downward
(3) $\frac{m_2 g}{m_1 + m_2}$, downward
(4) zero

22. A bullet of mass 0.02 kg is fired from a rifle horizontally into a 3 kg block of wood suspended by a string and gets embedded in the block. The impact causes the block to swing to a height of 20 cm above its initial level. The initial speed of the bullet is-

- (1) 302 m/s (2) 250 m/s
(3) 195 m/s (4) 350 m/s

23. A ball is moving with velocity 2 m/s towards a heavy wall moving towards the ball with speed 1 m/s as shown in figure. Assuming collision to be elastic, the velocity of ball immediately after collision is-

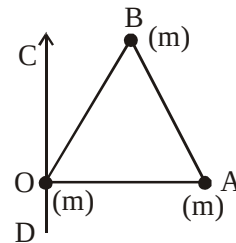


- (1) $4\hat{i}$ m/s (2) $2\hat{i}$ m/s
(3) $-4\hat{i}$ m/s (4) $-2\hat{i}$ m/s

24. Two particles of masses m_1 and m_2 are connected by a rigid massless rod of length r to constitute a dumb-bell which is free to move in the plane. The moment of inertia of the dumb-bell about an axis perpendicular of the plane passing through the centre of mass is :-

- (1) $\frac{m_1 m_2 r^2}{(m_1 + m_2)}$ (2) $(m_1 + m_2)r^2$
(3) $\frac{m_1 m_2 r^2}{(m_1 - m_2)}$ (4) $(m_1 - m_2)r^2$

25. Three point masses m are placed at the vertices of an equilateral triangle of side a . Moment of inertia of the system about an axis COD passing through a mass m at O and lying in the plane of AOB and perpendicular to OA is:-



- (1) $2ma^2$ (2) $\frac{2}{3}ma^2$ (3) $\frac{5}{4}ma^2$ (4) $\frac{7}{4}ma^2$

26. If a rigid body rolls on a surface without slipping, then :-

- (1) Angular speed is different at different point of a rigid body
(2) Linear speed is same at all points of the rigid body
(3) Linear speed is minimum at the highest point but maximum at the point of contact
(4) Linear speed is maximum at the highest point but minimum at the point of contact

27. A rigid body rotates about a fixed axis with variable angular velocity equal to $\alpha - \beta t$ at time t , where α and β are constants. The angle through which it rotates before it comes to rest is:-

- (1) $\frac{\alpha^2}{2\beta}$ (2) $\frac{\alpha^2 - \beta^2}{2\alpha}$ (3) $\frac{\alpha^2 - \beta^2}{2\beta}$ (4) $\frac{\alpha(\alpha - \beta)}{2}$

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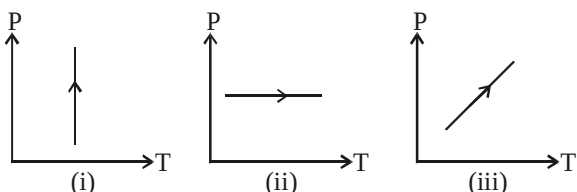
28. Two cylinders of equal size are filled with equal amount of ideal diatomic gas at room temperature. Both the cylinder are fitted with piston. In cylinder A, the piston is free to move while in B piston position is fixed. When same amount of heat is supplied to both the cylinders, the temperature of gas in A raises by 30 K. What will be the rise in temperature of gas in cylinder B ?

(1) 42K (2) 40K (3) 30K (4) 56K

29. If the internal energy of n_1 moles of He at 10 T temperature is equal to the internal energy of n_2 moles of H_2 at 6 T temperature. The ratio of (n_1/n_2) is :-

(1) $\frac{3}{5}$ (2) 2 (3) 1 (4) $\frac{5}{3}$

30. Pressure versus temperature graphs of an ideal gas are as shown in figure. Choose the wrong statement :-



- (1) Density of gas is increasing in graph (i)
(2) Density of gas is decreasing in graph (ii)
(3) Density of gas is constant in graph (iii)
(4) None of the above

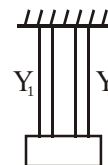
31. Two gases A and B having same pressure P, volume V and temperature T are mixed. If mixture has volume and temperature as V and T respectively, then the pressure of the mixture will be :-

(1) 4 P (2) 3 P (3) 2 P (4) P

32. If the volume of the gas containing n number of molecules is V, then pressure will decrease due to force of intermolecular attraction in the proportion:-

(1) $\frac{n}{V}$ (2) $\frac{n}{V^2}$
(3) $\left(\frac{n}{V}\right)^2$ (4) $\left(\frac{1}{V^2}\right)$

33. Two wires of same length and radius are joined in parallel and loaded. If the young's moduli of the materials of the wires are Y_1 and Y_2 and the combination behaves as a single wire then equivalent young's modulus is :-



(1) $Y = \frac{2Y_1Y_2}{Y_1 + Y_2}$ (2) $\frac{Y_1 + Y_2}{2}$

(3) $\frac{1}{Y} = \frac{1}{Y_1} + \frac{1}{Y_2}$ (4) $Y = \sqrt{Y_1Y_2}$

34. The reading of the pressure meter attached with a closed water pipe is $3.5 \times 10^5 \text{ Nm}^{-2}$. On opening the valve of the pipe, the reading of the pressure meter is reduced to $3 \times 10^5 \text{ Nm}^{-2}$. The speed of water flowing out of pipe will be :-

(1) 2 ms^{-1} (2) 5 ms^{-1}
(3) 8 ms^{-1} (4) 10 ms^{-1}

35. The workdone in blowing a soap bubble from an initial surface area of 0.5 cm^2 to a final surface area of 1.1 cm^2 , will be [surface tension of soap solution is $3 \times 10^{-2} \text{ Nm}^{-1}$] :-

(1) $18 \times 10^{-7} \text{ J}$ (2) $36 \times 10^{-7} \text{ J}$
(3) $90 \times 10^{-7} \text{ J}$ (4) $72 \times 10^{-7} \text{ J}$

36. A spherical ball of radius $1 \times 10^{-4} \text{ m}$ and density 10^4 kg/m^3 falls freely under gravity through a distance h, before entering a tank of water. If after entering the water, the velocity of ball does not change, the value of h is

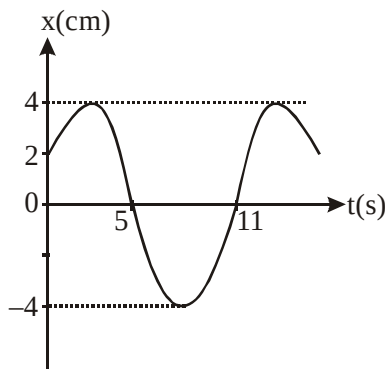
[viscosity of water = $9.8 \times 10^{-6} \text{ N s/m}^2$] :-

(1) 10.2 m (2) 20.4 m
(3) 80.8 m (4) 100.5 m

37. A particle executes simple harmonic motion according to equation $4 \frac{d^2x}{dt^2} + 320x = 0$. Its time period of oscillation is :-

- (1) $\frac{2\pi}{5\sqrt{3}}s$ (2) $\frac{\pi}{3\sqrt{2}}s$
(3) $\frac{\pi}{2\sqrt{5}}s$ (4) $\frac{2\pi}{\sqrt{3}}s$

38. Figure shows the position-time graph of an object in SHM. The correct equation representing this motion is :-



- (1) $2\sin\left(\frac{2\pi}{5}t + \frac{\pi}{6}\right)$ (2) $4\sin\left(\frac{\pi}{5}t + \frac{\pi}{6}\right)$
(3) $4\sin\left(\frac{\pi}{6}t + \frac{\pi}{3}\right)$ (4) $4\sin\left(\frac{\pi}{6}t + \frac{\pi}{6}\right)$

39. The potential energy of a particle executing SHM changes from maximum to minimum in 5 s. Then the time period of SHM is :-

- (1) 5 s (2) 10 s
(3) 15 s (4) 20 s

40. The earth's radius is R and acceleration due to gravity at its surface is g. If a body of mass m is sent to a height of $\frac{R}{4}$ from the earth's surface, the potential energy increases by :-

- (1) $mg\frac{R}{3}$ (2) $mg\frac{R}{4}$
(3) $mg\frac{R}{5}$ (4) $3mg\frac{R}{16}$

41. The orbit of a geostationary satellite is circular, the time period of satellite depends on :-

- I. mass of the satellite.
II. mass of the earth.
III. radius of the orbit.
IV. height of the satellite from surface of the earth.
Which of the following option is correct ?

- (1) Only I (2) I and II
(3) I, II and III (4) II, III and IV

42. Three waves of equal frequency having amplitudes 10 mm, 4 mm and 7 mm arrive at a given point with successive phase difference of $\pi/2$. The amplitude of the resulting wave in mm is :-

- (1) 7 (2) 6 (3) 5 (4) 4

43. Three similar wires of frequency n_1 , n_2 and n_3 are joined to make one wire. Its frequency will be :-

- (1) $n = n_1 + n_2 + n_3$
(2) $\frac{1}{n} = \frac{1}{n_1} + \frac{1}{n_2} + \frac{1}{n_3}$
(3) $\frac{1}{\sqrt{n}} = \frac{1}{\sqrt{n_1}} + \frac{1}{\sqrt{n_2}} + \frac{1}{\sqrt{n_3}}$
(4) $\frac{1}{n^2} = \frac{1}{n_1^2} + \frac{1}{n_2^2} + \frac{1}{n_3^2}$

44. In a hall, a person receives direct sound waves from a source 120 m away. He also receives waves from the same source which reach him after being reflected from the 25 m high ceiling at a point halfway between them. The two waves interfere constructively for wavelengths (in metres) :-

- (1) 10, $\frac{10}{2}$, $\frac{10}{3}$ (2) 20, $\frac{20}{3}$, $\frac{20}{5}$
(3) 30, 20, 10,... (4) 35, 25, 15,...

45. A tuning fork of frequency 90 Hz is sounded and moved towards an observer with a speed

equal to $\frac{1}{10}$ the speed of sound. The note heard

by the observer will have a frequency :-

- (1) 100 Hz (2) 110 Hz
(3) 80 Hz (4) 70 Hz

Use stop, look and go method in reading the question

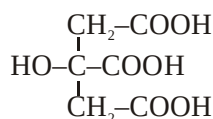
46. Which of the following statement is incorrect:-
 (1) 1st IP of N is greater than P
 (2) He < O < F size
 (3) $F + N^- \rightarrow F^- + N$ exothermic
 (4) $O + 2e^- \rightarrow O^{2-}$ endothermic
47. Select incorrect order(s) of electronegativity of element is/are :-
 (1) E.N. of F-atom on pauling scale < E.N. of F-atom on muliken scale
 (2) E.N. of Cl-atom in Cl_2O_7 > E.N. of Cl-atom in Cl_2O_5
 (3) E.N. of C-atom in CH_4 > E.N. of C-atom in CO_2
 (4) E.N. of Cu^{2+} > E.N. of Cu^+
48. If aufbau rule had not be followed then Ca would be placed in :-
 (1) s-block (2) p-block
 (3) d-block (4) f-block
49. Which of the following statement regarding orthoboric acid (H_3BO_3) is FALSE.
 (1) It is a weak monobasic acid
 (2) Soluble in hot water
 (3) It has planar structure
 (4) It acts as tribasic acid
50. Which of the following does not exist in free form :-
 (1) BF_3 (2) BBr_3
 (3) BH_3 (4) BI_3
51. Which of the following equation is not correctly matched :-
 (1) H_3BO_3 is a weak mono basic acid as it liberates hydroges ions as

$$H_3BO_3 \longrightarrow H^+ + H_2BO_3^-$$

 (2) $H_3BO_3 \xrightarrow{\Delta} HBO_2 \xrightarrow[\text{Boric oxide}]{\text{Red hot}} B_2O_3$
 (3) $2BN + 6H_2O \rightarrow 2H_3BO_3 + 2NH_3$
 (4) $Na_2B_4O_7 \cdot 10H_2O + 2HCl \rightarrow 2NaCl + 4H_3BO_3 + 5H_2O$
52. What will be the value of formal charge on individual atom of O_3 molecule :-
 (1) +1, +1, +1 (2) 0, +1, -1
 (3) 0, 0, 0 (4) +2, -2, 0
53. In Cl_2O_7 the π -bond between Cl-O is formed by the over lapping of :-
 (1) sp^3-2p (2) $2p-3p$
 (3) sp^3-sp^3 (4) $2p-3d$
54. Which of the following have been arranged in increasing order of stability :-
 (1) $O_2^{-2} < O_2^- < O_2^+ < O_2$
 (2) $O_2^{-2} < O_2^- < O_2 < O_2^+$
 (3) $O_2 < O_2^+ < O_2^{-2} < O_2^-$
 (4) $O_2^+ < O_2^{2-} < O_2^- < O_2$
55. The vapor pressure of o-nitrophenol at any given temperature is predicted to be-
 (1) higher than that of p-nitrophenol
 (2) lower than that of p-nitrophenol
 (3) same as that of p-nitrophenol
 (4) higher or lower depending upon the size of the vessel
56. Which of the following is strongest attraction ?
 (1) $Cl^- - (H-H)$ (2) $CHCl_3 - (CHCl_3)$
 (3) $CCl_4 - (H_2O)$ (4) $Cl^- - (H_2O)$
57. Which of the following silicate has no neutral covalently bonded oxygen ?
 (1) Pyrosilicate (2) Orthosilicate
 (3) Chain silicate (4) Sheet silicate
58. When Li_2CO_3 (A) & $Na_2CO_3 \cdot 10H_2O$ (B) is heated loss of weight is observed this loss is due to :-
 (1) A (2) B
 (3) Evaporation of NaOH (4) both A & B
59. Polarisability of halide ions increases in the order ?
 (1) F^-, I^-, Br^-, Cl^- (2) Cl^-, Br^-, I^-, F^-
 (3) I^-, Br^-, Cl^-, F^- (4) F^-, Cl^-, Br^-, I^-
60. Intermolecular forces in solid hydrogen are :-
 (1) Covalent forces (2) Vanderwaal forces
 (3) Hydrogen bonds (4) All

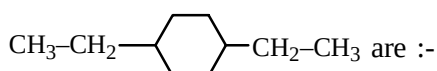
(Take it Easy and Make it Easy)

61. IUPAC name of give compound is



- (1) 3-Carboxy-3-hydroxy pentanedicarboxylic acid
(2) 2-Hydroxy propane-1,2,3-tricarboxylic acid
(3) 2-Hydroxy propane-1,2,3-trioic acid
(4) 3-Hydroxy propane-1,2,3-tricarboxylic acid

62. $\text{CH}_3\text{--}$  $\text{--CH}_2\text{--CH}_2\text{--CH}_3$ and

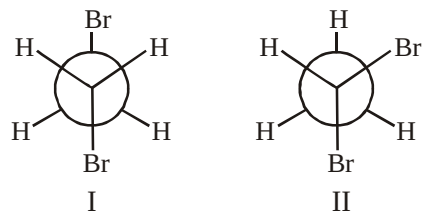


- (1) Chain isomer (2) Position isomer
(3) Metamer (4) Functional isomer

63. Minimum carbon required for ester to show metamerism :-

- (1) 2 (2) 3 (3) 4 (4) 5

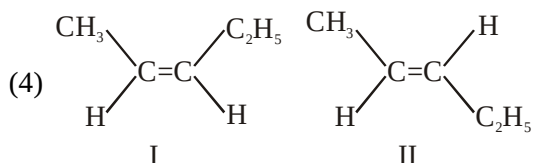
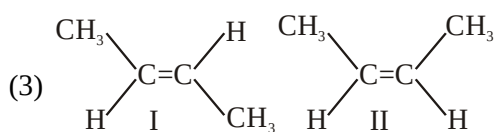
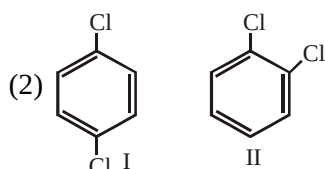
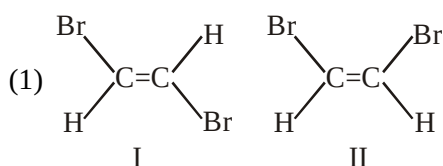
64.



The structure I & II are related with each other as :-

- (1) enantiomer (2) conformer
(3) position isomer (4) geometrical isomer

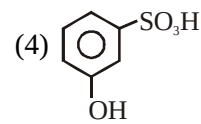
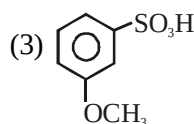
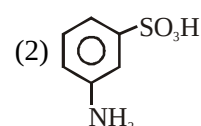
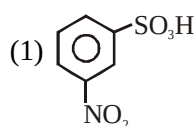
65. In which of the following I have higher dipole moment than II :-



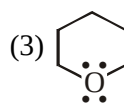
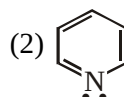
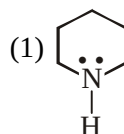
66. $\begin{array}{c} \text{H} \\ \diagdown \\ \text{C}=\text{C} \\ \diagup \\ \text{CH}_3 \end{array}$ (A) which one is correct :-
(B)

- (1) If (A) & (B) have same molecular formula then compound never show G.I.
(2) If (A) & (B) have same molecular formula than compound always show G.I.
(3) If (A) & (B) have same molecular formula, then compound may or may not show G.I.
(4) None of the above

67. Which of the following is strongest acid?

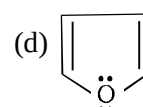
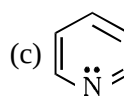
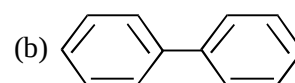
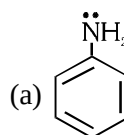


68. Identify the strongest base from the given compounds :-



(4) All have same basic strength

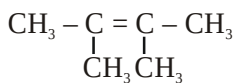
69. Which of the following compounds are aromatic in nature :-



- (1) a, c
(3) a, c, d

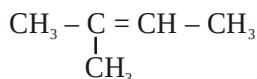
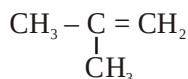
- (2) c, d
(4) a, b, c, d

70. Stability of following alkenes in the increasing order is :-



(I)

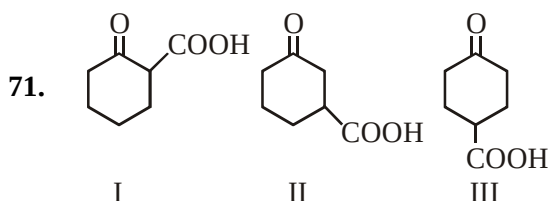
(II)



(III)

(IV)

- (1) $I < III < IV < II$
 (2) $I < II < III < IV$
 (3) $IV < III < II < I$
 (4) $III > I > IV > II$

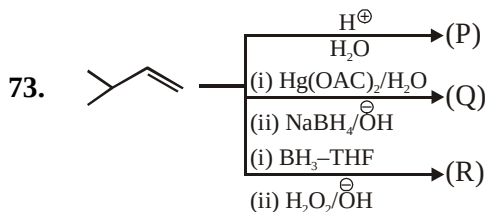


Which of the following order is correct for the ease of decarboxylation of these carboxylic acids?

- (1) $I > II > III$
- (2) $III > II > I$
- (3) $II > I > III$
- (4) $III > I > II$

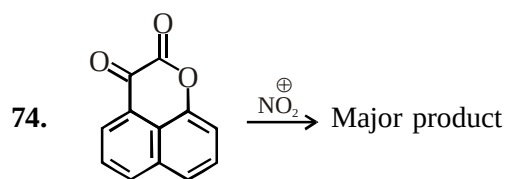
72. In which of the reaction antimarkovnikoff product is form :

- (1) $\text{CF}_3\text{CH}=\text{CH}_2 \xrightarrow{\text{HCl}}$
 (2) $\text{ClCH}=\text{CH}_2 \xrightarrow{\text{HCl}}$
 (3) $\text{CH}_3\text{OCH}=\text{CH}_2 \xrightarrow{\text{HCl}}$
 (4) None

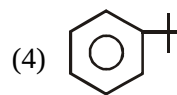
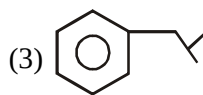
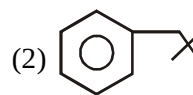
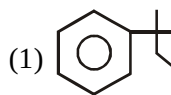
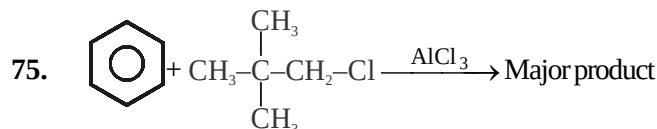
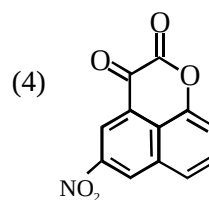
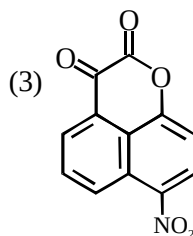
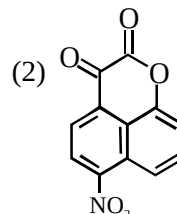
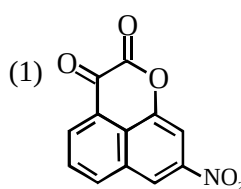


Choose the correct option ?

- (1) (P) & (Q) $\longrightarrow$ Chain isomers
- (2) (P) & (R) $\longrightarrow$ Position isomers
- (3) (Q) & (R) $\longrightarrow$ Identical compound
- (4) (P) & (Q) $\longrightarrow$ Functional isomers



Find the major product :-



76. $2\text{KClO}_3 \longrightarrow 2\text{KCl} + 3\text{O}_2$

is an example of

- (1) Intermolecular redox reaction
- (2) Intramolecular redox reaction
- (3) Disproportionation redox reaction
- (4) None

77. If 30 ml of H_2 and 20 ml of O_2 reacts to form water, what is left at the end of the reaction if all volume measured at same conditions of pressure and temperature :-

- (1) 10 ml of H_2 (2) 5 ml of H_2
(3) 10 ml of O_2 (4) 5 ml of O_2

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78. The standard Gibb's free energy change, ΔG° is related to equilibrium constant K_p as :-

(1) $K_p = -RT \ln \Delta G^\circ$ (2) $K_p = \left(\frac{e}{RT} \right) \Delta G^\circ$

(3) $K_p = \frac{-\Delta G^\circ}{RT}$ (4) $K_p = e^{\frac{\Delta G^\circ}{RT}}$

79. Why are strong acids generally used as standard solutions in acid-base titrations.

- (1) The pH at the equivalence point will always be 7
(2) They can be used to titrate both strong and weak base
(3) Strong acids form more stable solution than weak acids
(4) The salts of strong acids do not hydrolysed

80. Choose the pair of species in which Oxidation Number of nitrogen is same :-

- (1) NO_2^- , N_2O_3 (2) NO_3^- , NO_2
(3) NO , N_2O_5 (4) None

81. The frequency of radiation emitted when the electron falls from $n = 4$ to $n = 1$ in a H-atom will be : (Given ionization energy of H = $2.18 \times 10^{-18} \text{ J atom}^{-1}$ and $h = 6.625 \times 10^{-34} \text{ Js}$):-

- (1) $3.07 \times 10^{15} \text{ s}^{-1}$ (2) $2.00 \times 10^{15} \text{ s}^{-1}$
(3) $1.54 \times 10^{15} \text{ s}^{-1}$ (4) $1.03 \times 10^{15} \text{ s}^{-1}$

82. When inert gas is added at constant volume in the reaction $2\text{NOCl(g)} \rightleftharpoons 2\text{NO(g)} + \text{Cl}_2\text{(g)}$ at equilibrium the amount of NO :-

- (1) Increases (2) Decreases
(3) Remains same (4) None of these

83. $\text{PCl}_{5\text{(g)}} \rightleftharpoons \text{PCl}_{3\text{(g)}} + \text{Cl}_{2\text{(g)}}$ Initially pressure is 6 atm and total pressure at equilibrium is 10 atm. Then find out K_p :-

- (1) 1 atm (2) 8 atm
(3) 16 atm (4) 4 atm

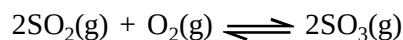
84. The value of the energy of electron for the first excited state of hydrogen atom will be :-

- (1) -13.6 eV (2) -3.40 eV
(3) -1.51 eV (4) -0.85 eV

85. Equivalent mass of a metal is double than that of oxygen, 1.6 g of the metal on reation with acid displaces :-

- (1) 0.1 mol H_2
(2) 0.1 equivalent H_2
(3) 2.24 mL H_2 at STP
(4) 0.2 mol H_2

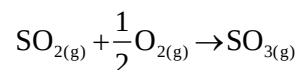
86. A mixture of SO_3 , SO_2 and O_2 gases is maintained in a 10 litre flask at the temperature at which the equilibrium constant for the reaction is 100;



If no. of moles of SO_2 and SO_3 in the flask are equal, at equilibrium how many moles of O_2 are present ?

- (1) 0.4 mol (2) 0.01 mol
(3) 0.1 mol (4) 0.04 mol

87. The values of heat of formation of SO_2 and SO_3 are -298.2 kJ and -98.2 kJ. The heat of reaction of the following reaction will be :



- (1) -200 kJ (2) -356.2 kJ
(3) +200 kJ (4) -396.2 kJ

88. 24 L of H_2 and 22.4 L of Cl_2 are mixed and exploded. The composition by volume of mixture is :-

- (1) 0.8 L of $\text{H}_2\text{(g)}$ and 22.4 L HCl(g)
(2) 22.4 L HCl(g)
(3) 1.6 L $\text{Cl}_2\text{(g)}$ and 44.8 L HCl(g)
(4) 1.6 L $\text{H}_2\text{(g)}$ and 44.8 L HCl(g)

89. In which of the following the oxidation number of oxygen has been arranged in increasing order:-

- (1) $\text{OF}_2 < \text{KO}_2 < \text{BaO}_2 < \text{O}_3$
(2) $\text{BaO}_2 < \text{KO}_2 < \text{O}_3 < \text{OF}_2$
(3) $\text{BaO}_2 < \text{O}_3 < \text{OF}_2 < \text{KO}_2$
(4) $\text{KO}_2 < \text{OF}_2 < \text{O}_3 < \text{BaO}_2$

90. At 300 K, the density of a certain gaseous molecule at 2 bar is double to that of dinitrogen (N_2) at 4 bar. The molar mass of gaseous molecule is :-

- (1) 28 g mol^{-1} (2) 56 g mol^{-1}
(3) 224 g mol^{-1} (4) 112 g mol^{-1}

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91. In binomial nomenclature system, when name of any plant is hand written then it should be :-
 (1) Italicized
 (2) Underlined cummulatively
 (3) Underlined separately
 (4) Closed in bracket
92. Which of the following is included in kingdom Protista according to five kingdom system :-
 (1) Mycoplasma
 (2) Protozoans
 (3) Blue green algae
 (4) *Agaricus*
93. Which step is not involve in fungal sexual reproduction :-
 (1) Plasmogamy
 (2) Meiosis
 (3) Embryo formation
 (4) Karyogamy
94. Which of the following fungi is used extensively in biochemical and genetic work ?
 (1) *Aspergillus*
 (2) *Claviceps*
 (3) *Agaricus*
 (4) *Neurospora*
95. Whittaker failed to give any place to one of the following in his classification :-
 (1) Cyanobacteria
 (2) Virus
 (3) *Euglena*
 (4) *Chlamydomonas*
96. Choose the correct set of bacterial disease :-
 (1) Mumps, Cholera, Dengue
 (2) Chicken pox, Typhoid, Mumps
 (3) Mumps, Tetanus, Chicken pox
 (4) Cholera, Typhoid, Tetanus
97. Indestructible cell wall found in which protist :-
 (1) Dinoflagellates (2) Chrysophytes
 (3) Euglenoids (4) Slime mould
98. What is not found in members of deuteromycetes :-
 (1) Asexual reproduction
 (2) Branched and septated mycelium
 (3) Sexual reproduction
 (4) Conidia formation
99. In which of the following group there is no independent free living existence of gametophyte:-
 (1) Bryophyta
 (2) Pteridophyta
 (3) Gymnosperm
 (4) Thallophyta
100. Choose the incorrect match from the following:-
 (1) Red algae - Carrageen
 (2) Green algae - Pyrenoids
 (3) Brown algae - Alginates
 (4) Blue green algae - Agar-Agar
101. Which of the following is last stage of sporophyte generation in pteridophytes :-
 (1) Spore
 (2) Embryo
 (3) Zygote
 (4) Spore mother cell
102. Which structure is formed in the life cycle of moss, between spore and main plant :-
 (1) Prothallus
 (2) Embryo
 (3) Spore mother cell
 (4) Protonema
103. Which option is not related with gametophyte of Pteridophytes :-
 (1) Sex organ
 (2) Vascular tissue
 (3) Gamete
 (4) Photosynthetic ability
104. Pteridophyta differ from bryophyta in having :-
 (1) Independent gametophyte
 (2) Independent sporophyte
 (3) Dependent gametophyte
 (4) Dependent sporophyte

105. Which one out of A-D given below correctly represents the structural formula of the acidic amino acid :-

A	B	C	D
$\begin{array}{c} \text{NH}_2 \\ \\ \text{H}-\text{C}-\text{COOH} \\ \\ \text{CH}_2-\text{OH} \end{array}$	$\begin{array}{c} \text{NH}_2 \\ \\ \text{H}-\text{C}-\text{COOH} \\ \\ \text{CH}_2 \\ \\ \text{CH}_2 \\ \\ \text{CH}_2 \\ \\ \text{CH}_2 \\ \\ \text{NH}_2 \end{array}$	$\begin{array}{c} \text{NH}_2 \\ \\ \text{H}-\text{C}-\text{COOH} \\ \\ \text{CH}_2 \\ \\ \text{CH}_2 \\ \\ \text{COOH} \end{array}$	$\begin{array}{c} \text{NH}_2 \\ \\ \text{H}-\text{C}-\text{COOH} \\ \\ \text{CH}_2 \\ \\ \text{C}_6\text{H}_{11}\text{OH} \end{array}$

Options :-

- (1) D (2) C (3) B (4) A

106. Sucrose is made up of :-
 (1) α -D glucose and β -D glucose
 (2) α -D glucose and β -D fructose
 (3) β -D glucose and α -D fructose
 (4) β -D glucose and β -D galactose
107. Match the column I with column II and choose correct option :-

	Column-I		Column-II
i	Lectins	a	Codeine
ii	Essential oil	b	Lemon grass
iii	Pigment	c	Concanavalin A
iv	Alkaloid	d	Anthocyanin

	i	ii	iii	iv
(1)	d	c	b	a
(2)	c	b	d	a
(3)	a	b	d	c
(4)	b	c	d	a

108. Which of the following polymer is produced by polymerisation of N-acetyl glucosamine :-
 (1) Chitin (2) Cellulose
 (3) Agar-agar (4) Lipid

109. Which of the following statement is not incorrect :-

- (1) Lipid is glycerol which is trihydroxyhexane
 (2) Oils have higher melting point hence remain as oil in winter
 (3) Arachidonic acid has 20 carbon atoms including the carboxyl carbon
 (4) Cell wall is made up of phospholipid in which lipids have phosphorus and a phosphorylated organic compound

110. Find out the miss-matched pair related to cockroach ?

- (1) Wings - 2 Pairs
 (2) Legs- 3 Pairs
 (3) Anal cerci- 1 Pair
 (4) Antennae- 2 Pairs

111. Identify the odd one related to circulatory system of earthworm ?

- (1) Lateral hearts
 (2) Subneural blood vessel
 (3) Anterior loops
 (4) Erythrocytes

112. Find the incorrect one related to *Pleurobrachia*?

- (1) Comb plates
 (2) Bioluminescence
 (3) External fertilisation
 (4) Triploblastic

113. In which of the following animal body is covered by a calcareous shell and is unsegmented with a distinct head, muscular foot and visceral hump ?

- (1) *Asterias*
 (2) *Saccoglossus*
 (3) *Pila*
 (4) *Ascidia*

114. Which of the following animal have direct development ?

- (1) *Nereis*
 (2) *Anopheles*
 (3) *Octopus*
 (4) *Rana*

115. Which of the following fish is without placoid scales ?

- (1) *Pristis* (2) *Scoliodon*
(3) *Carcharodon* (4) *Catla*

116. True coelom first developed in which phylum ?

- (1) Aschelminthes (2) Annelida
(3) Arthropoda (4) Chordata

117. All of the following animals show metagenesis except :-

- (1) *Obelia* (2) *Physalia*
(3) *Hydra* (4) *Porpita*

118. Match the following columns and choose the correct answer :-

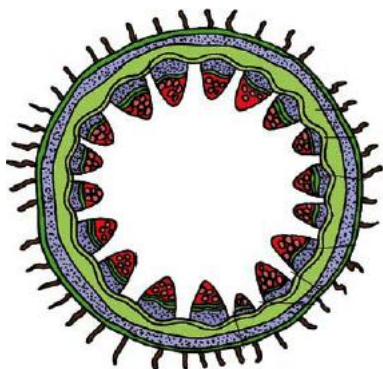
	Column-I (Animals)		Column-II (Structure)
A	<i>Taenia</i>	I	Cnidocytes
B	<i>Culex</i>	II	Calcareous ossicles
C	<i>Echinus</i>	III	Malpighian tubules
D	<i>Pennatula</i>	IV	Flame cells

- (1) A-II, B-III, C-IV, D-I
(2) A-IV, B-III, C-I, D-II
(3) A-IV, B-III, C-II, D-I
(4) A-I, B-II, C-III, D-IV

119. Bicollateral vascular bundles are found in members of :-

- (1) Cucurbitaceae (2) Poaceae
(3) Liliaceae (4) Malvaceae

120. Given figure is of



- (1) Monocot stem (2) Dicot stem
(3) Monocot root (4) Dicot root

121. Spring wood is :-

- (1) Lighter in colour and lower in density
(2) Lighter in colour and higher in density
(3) Darker in colour and higher in density
(4) Darker in colour and lower in density

122. Meristem occurring in grasses that regenerate parts removed by the grazing herbivores is :-

- (1) Apical meristem
(2) Intercalary meristem
(3) Secondary meristem
(4) Lateral meristem

123. The inflorescence present in *Euphorbia* is

- (1) Cyathium
(2) Capitulum
(3) Helicoid cymose
(4) Verticillaster

124. Stolon is found in :-

- (1) *Eichornia*
(2) *Pistia*
(3) *Fragaria*
(4) All

125. Edible part of lemon is :-

- (1) Epicarp
(2) Mesocarp
(3) Glandular hairs
(4) Endocarp

126. Floral formula of solanaceae is :-

- (1) $\text{Br } \frac{\text{♂}}{\text{♀}} \text{K}_{(5)} \text{C}_5 \text{A}_5 \underline{\text{G}_{(2)}}$
(2) $\text{Br } \oplus \frac{\text{♀}}{\text{♂}} \text{K}_{(5)} \text{C}_{(5)} \text{A}_5 \underline{\text{G}_{(2)}}$
(3) $\text{Br } \oplus \frac{\text{♂}}{\text{♀}} \text{K}_{(5)} \text{C}_{(5)} \text{A}_5 \underline{\text{G}_{(2)}}$
(4) $\text{Br } \oplus \frac{\text{♂}}{\text{♀}} \text{K}_{(5)} \text{C}_{(5)} \text{A}_3 \underline{\text{G}_{(2)}}$

127. Asparagus is member of :-

- (1) Poaceae
(2) Liliaceae
(3) Solanaceae
(4) Asteraceae

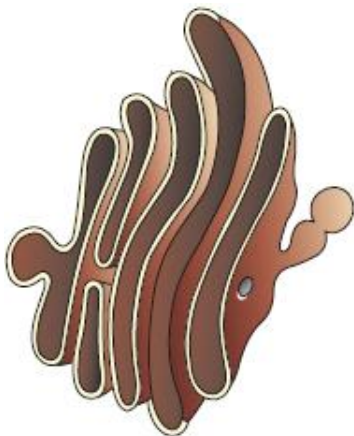
128. A-Peroxisome B-Mitochondria
C-Nucleolus D-Centrosome
E-Nucleus F-Sphaerosome
Choose the organelles from above (A to F) which bound by single membrane in animal cell :-
(1) A, B, C, E, F (2) A, C, D, E
(3) A only (4) A, C, E, F

129. In a plant cell :-
(1) Central vacuole is not present
(2) Centrioles are not present for cell division
(3) Semi fluid cell membrane is not present
(4) Generally lysosomes are present for the Intracellular digestion of foreign materials of Endocytosis

130. Which one is correctly Matched :-
(1) Amyloplasts- Store Proteins
(2) Elaioplast- Contain pigments
(3) Chloroplast- Carotenoids Present
(4) Chromoplast- Water soluble pigments are present

131. Choose the correct statement for chromosome :-
(1) In telocentric chromosome one arm is very long and one is very short
(2) Acrocentric-one arm is very short
(3) Metacentric-one arm is very long
(4) Submetacentric-Centromere found in the middle of the chromosome

132.



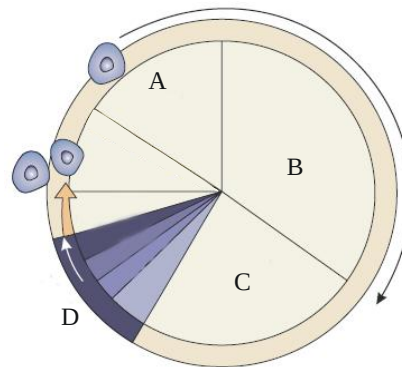
- Choose the correct option for given figure :-
(1) Site for active RNA synthesis
(2) Main site for Protein synthesis
(3) Important site for the formation of glycoproteins and glycolipids
(4) Site for the digestion of macromolecules

133. Cilia, Flagella and centrioles are mainly made up of :-

- (1) Intermediate filaments
(2) Phospholipid bilayer
(3) Microtubules
(4) Microfilaments

134. A-Condensation of chromatin materials
B-Nuclear envelope reform
C-Attachment of spindle fibres to kinetochore
D-Movement of sister chromatids toward poles
Choose the correct sequence of events given above (AtoD) for mitosis :-
(1) A→B→C→D (2) A→C→D→B
(3) A→C→B→D (4) A→D→C→B

135.



Choose the incorrect option for A to D in given diagram :-

- (1) A- Cytoplasmic growth
(2) B- Chromosome number double
(3) C- Tubulin protein synthesis
(4) D- Dramatic phase

136. Recombinase activity occurs in :-

- (1) Prophase-I
(2) Prophase
(3) Metaphase-I
(4) Telophase-I

137. Which of the following organisms retain the enzymatic machinery to partially oxidise glucose to pyruvic acid ?

- (1) Aerobic organisms
(2) Anaerobic organisms
(3) Most of the anaerobic and some aerobic organisms.
(4) All the living organisms

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138. In mitochondrial ETS, cytochrome 'C' acts as a mobile carrier for transfer of electrons between :-

- (1) Complex III and IV
- (2) Complex I and II
- (3) Complex II and III
- (4) Complex IV and V

139. *Rhizobium* has symbiotic relationship with the roots of :-

- (1) Garden pea and *Alnus*
- (2) *Alfalfa* and *Alnus*
- (3) Sweet clover and *Alnus*
- (4) Sweet pea and Lentils

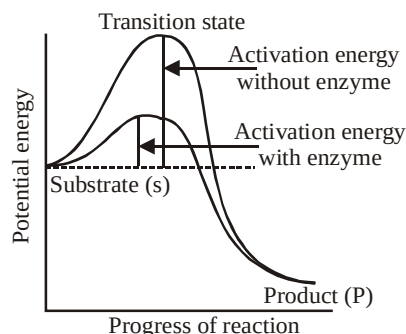
140. Low rate of respiration in host cells may affect the process of symbiotic nitrogen fixation because-

- (1) it highly activates the nitrogen fixing microbes
- (2) production of leghaemoglobin requires the higher rate of respiration
- (3) it may cause the inactivation of oxygen scavenger required for nitrogen fixation
- (4) the energy required for symbiotic nitrogen fixation is obtained from the respiration of the host cells

141. Which of the following is **not correct** ?

- (1) A plant cell show plasmolysis when it is placed in a solution which has lower water potential as compare to the protoplasm of cell
- (2) Imbibition is active transport of water since water movement in imbibition is against the water potential gradient
- (3) Water potential gradient and affinity between the solid and liquid is must for imbibition
- (4) The pressure exerted by protoplast against the cell wall is responsible for enlargement of cells

142.



Above diagram showing the concept of activation energy. If in this diagram energy level of 'P' become higher than the energy level of 'S' then which of the following will **not** apply for this diagram ?

- (1) The process will be endothermic
- (2) Activation energy with enzyme will be less than activation energy without enzyme
- (3) Energy difference between transition state and product will be more than activation energy
- (4) The substrate will go through the transition state

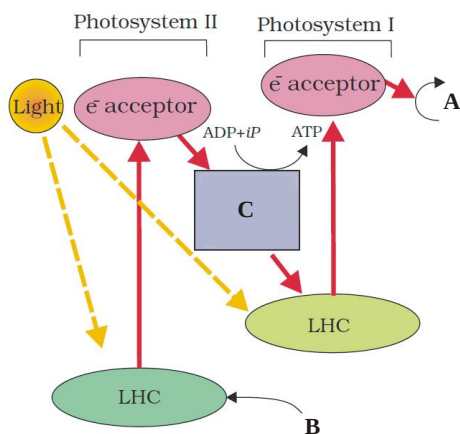
143. If we cut the stem of a small soft stemmed plant horizontally near the base, in the morning, drops of solution may ooze out of the stem. This happens due to :-

- (1) Guttation
- (2) Root pressure
- (3) Transpiration pull
- (4) Plasmolysis

144. Which of the following plant growth regulators, is derivative of accessory photosynthetic pigments?

- (1) Absciscic acid
- (2) Gibberellic acid
- (3) Indole-3-acetic acid
- (4) Ethylene

145. Carbondioxide is essential for photosynthesis, this fact can be proved by using :-
 (1) KOH, a bell jar and a candle.
 (2) A test tube or a jar, KOH and a leafy plant.
 (3) KOH and purple sulphur bacteria.
 (4) A bell jar, candle and a mouse.
146. One molecule of enzyme converts the 5 molecules of substrate into product in 5 minutes. 20 molecules of substrate and two molecules of enzyme are mixed in a test tube. After 10 minutes the test tube will have :-
 (1) products and 10 unreacted molecules of substrate
 (2) only products
 (3) 10 unreacted molecules of substrate, enzymes and products
 (4) products and enzymes
147. Line diagram represented below is the Z scheme of light reaction in photosynthesis. Which one of the following options is **correct** for this process ?



- (1) This process occurs in the stroma lamellae as well as in the grana lamellae
 (2) In the part 'C' of this scheme the movement of electrons is uphill in terms of redox potential scale
 (3) Process 'B' of this scheme results in the release of one of the net products of photosynthesis
 (4) Process 'A' of this scheme occurs on the inner side of the grana thylakoid

148. (i) Required in abundant quantities in meristem tissue
 (ii) Helps to maintain cation-anion balance in cells
 (iii) Maintenance of the turgidity of cells
 Above functions are related to which of the following essential elements ?
 (1) Magnesium
 (2) Potassium
 (3) Molybdenum
 (4) Chlorine
149. In plants what appears as symptoms of manganese toxicity may actually be the deficiency symptoms of :-
 (1) iron, boron and calcium
 (2) magnesium, zinc and copper
 (3) calcium, sulphur and nitrogen
 (4) iron, magnesium and calcium
150. The type of growth, in which only one daughter cell among the two (produced by the division of meristem cell) retain the ability to divide, shows which of the following growth curves ?
 (1) Linear curve
 (2) Sigmoid curve
 (3) J-shape curve
 (4) Bell shape curve
151. A special center which can moderate the cardiac function through autonomic nervous system is present in___ :-
 (1) Hypothalamous
 (2) Cerebrum
 (3) Cerebellum
 (4) Medulla oblongata
152. Blood grouping is based on the presence or absence of antigen. In Rh blood group Rh factor present on :-
 (1) RBCs surface
 (2) WBCs surface
 (3) Platelets surface
 (4) In Plasma

- | | |
|--|--|
| <p>153. In human heart Purkinje fibres present in :-
 (1) Throughout the ventricular musculature
 (2) Rt atria
 (3) Only in Inter Ventricle septum
 (4) Both Atria & Ventricle</p> <p>154. In a healthy individual approximately GFR is :-
 (1) 125 ml/hr
 (2) 180 litres/day
 (3) 180,000 ml/hr
 (4) 50 ml/day</p> <p>155. Malfunctioning of kidneys can lead to accumulation of urea in blood, a condition called:-
 (1) Uremia
 (2) Glomerulonephritis
 (3) Renal calculi
 (4) Ketonuria</p> <p>156. Antennal glands or green glands are the excretory structure of :-
 (1) Prawn
 (2) Cockroach
 (3) Earthworm
 (4) <i>Planaria</i></p> <p>157. Prolonged <u>A</u> leads to a complex disorder called <u>B</u>. Find the correct option which fills above blanks correctly :-
 (1) A-hypoglycemia, B-diabetes insipidus
 (2) A-hyperglycemia, B-diabetes insipidus
 (3) A-hypoglycemia, B-diabetes mellitus
 (4) A-hyperglycemia, B-diabetes mellitus</p> <p>158. Erythropoietin which stimulates erythropoiesis is secreted by :-
 (1) JG cells of heart
 (2) JG cells of kidney
 (3) JG cells of pancreas
 (4) JG cells of gastro-intestinal tract</p> | <p>159. Secretin acts on the :-
 (1) endocrine pancreas and stimulates secretion of water and bicarbonate ions
 (2) exocrine pancreas and stimulates secretion of water and bicarbonate ions
 (3) endocrine pancreas and stimulates secretion of insulin and glucagon
 (4) Exocrine pancreas and stimulates secretion of insulin and glucagon</p> <p>160. Select the odd group related to chemical nature of hormone :-
 (1) Insulin, glucagon
 (2) Gastrin, secretin
 (3) Cortisol, estradiol
 (4) Thymosin, Parathyroid hormone</p> <p>161. Which is a major coordinating centre for sensory and motor signaling ?
 (1) Cerebellum
 (2) Thalamus
 (3) Adenohypophysis
 (4) Epiphysis cerebri</p> <p>162. The Cerebral hemispheres are connected by a tract of nerve fibres called :-
 (1) Corpus luteum
 (2) Corpus albicans
 (3) Corpus callosum
 (4) Corpus cavernosum</p> <p>163. Corpora quadrigemina found in :-
 (1) Fore brain (2) Mid brain
 (3) Hind brain (4) Spinal cord</p> <p>164. Cardiovascular reflexes centre present in :-
 (1) Cerebrum
 (2) Pons
 (3) Medulla oblongata
 (4) Cerebellum</p> <p>165. Optimum pH for salivary amylase is :-
 (1) 6.8 (2) 1.8
 (3) 7.8 (4) 8.4</p> |
|--|--|

Time Management is Life Management

166. Chylomicrons are transported into :-

- (1) intestinal mucosa
- (2) Lacteals
- (3) Blood capillary
- (4) Mucosa of stomach

167. Dental formula of monophyodont teeth in human being is :-

- (1) $\frac{2123}{2123} \times 2$
- (2) $\frac{2102}{2102} \times 2$
- (3) $\frac{0021}{0021} \times 2$
- (4) $\frac{2120}{2120} \times 2$

168. Which factor converts chymotrypsinogen to chymotrypsin ?

- (1) HCl
- (2) Ca^{2+}
- (3) Trypsin
- (4) Pepsin

169. The amount of inspired or expired air per minute by a healthy man is :-

- (1) 500 ml
- (2) 2500-3000 ml
- (3) 1000-1100 ml
- (4) 6000-8000 ml

170. Dissociation of CO_2 from carbamino-haemoglobin in alveoli is due to :-

- (1) Low PO_2 and High PCO_2
- (2) Low PO_2 and low PCO_2
- (3) High PO_2 and High PCO_2
- (4) High PO_2 and low PCO_2

171. Cigarette smoking leads to lung cancer. This disorder of lungs which mainly occurs due to cigarette smoking is called :-

- (1) Asthma
- (2) Emphysema
- (3) Pneumonia
- (4) Bronchitis

172. Arrange the following steps of muscle contraction in the sequence of events occurring first :

- (A) Receptor sites on sarcolemma
- (B) Nerve impulse
- (C) Release of Ca^{++}
- (D) Acetylcholine release
- (E) Shortening of sarcomere
- (F) Synaptic cleft
- (G) Spread of impulse over sarcolemma on T-tubule

The correct option is :

- (1) $\text{B} \rightarrow \text{D} \rightarrow \text{F} \rightarrow \text{A} \rightarrow \text{G} \rightarrow \text{C} \rightarrow \text{E}$
- (2) $\text{B} \rightarrow \text{D} \rightarrow \text{A} \rightarrow \text{F} \rightarrow \text{G} \rightarrow \text{C} \rightarrow \text{E}$
- (3) $\text{B} \rightarrow \text{D} \rightarrow \text{A} \rightarrow \text{F} \rightarrow \text{G} \rightarrow \text{E} \rightarrow \text{C}$
- (4) $\text{D} \rightarrow \text{B} \rightarrow \text{A} \rightarrow \text{D} \rightarrow \text{G} \rightarrow \text{E} \rightarrow \text{C}$

173. The joint of our neck which allows us to rotate our head left to right is :

- (1) Pivot joint
- (2) Hinge joint
- (3) Saddle joint
- (4) Ellipsoid joint

174. Smooth muscles are:

- (1) Found in the walls of heart only
- (2) Found in the walls of all the hollow organs except heart
- (3) Attached to the bones only
- (4) Found in the walls of alimentary canal

175. Human ears perform sensory functions. These are :

- (1) Hearing organs
- (2) Maintenance of body balance
- (3) Both 1 and 2
- (4) Voice production

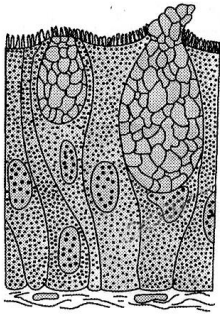
176. The adult human eyeball is nearly a _____ structure :

- (1) Oval
- (2) Circular
- (3) Opaque
- (4) Spherical

177. Which of the following tissue are modified for storage of fat ?

- (1) Areolar tissue
- (2) Cartilage
- (3) Muscular tissue
- (4) Adipose tissue

178.



Identify the structure shown in the diagram ?

- (1) Salivary glands (2) Mammary glands
(3) Gastric glands (4) Goblet cells

179. Cuboidal epithelium with microvilli (brush-bordered) is observed in ?

- (1) DCT
(2) Gall bladder
(3) PCT
(4) Vagina

180. Fibroblasts and collagen fibres oriented in a criss crossed pattern is found in ?

- (1) Dense irregular tissue
(2) Stratified columnar epithelium
(3) Adipose tissue
(4) Cartilage

Your moral duty is to prove that
ALLEN is ALLEN



◆ Your Hard Work Leads to Strong Foundation ◆